DSO 530: APPLIED MODERN STATISTICAL LEARNING METHODS

INSURANCE LOSS ANALYTICS

Predicting Loss Cost, Historically Adjusted Loss Cost, and Claim Status

GROUP 3 | USC Marshall School of Business

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Executive Summary

Insurance pricing depends on accurately predicting expected claim losses. Underpricing high-risk policyholders and overpricing low-risk ones both create adverse selection, eroding profitability. This project predicts Loss Cost per Exposure Unit (LC), Historically Adjusted Loss Cost (HALC), and Claim Status (CS) using 39,928 auto insurance policy records with 28 variables.

Target	Final Model	Validation Metric	Why selected
LC	Stacking Ensemble (LightGBM + XGBoost + Random Forest; Ridge meta-learner)	MSE = 316,465	Achieved the lowest validation MSE by combining complementary nonlinear learners, capturing both frequency and severity patterns under Tweedie loss.
HALC	Stacking Ensemble (LightGBM + XGBoost + Random Forest; Ridge meta-learner)	MSE = 1,005,405	Maintained the best predictive performance while preserving stability under historical adjustment.
CS	XGBoost Classifier	ROC-AUC = 0.7884	Provided the best discrimination under severe class imbalance with optimized scale_pos_weight.

The strongest predictive signals are behavioral and contractual rather than purely vehicle-based: policy duration, cancellation rate, and net premium dominate SHAP importance. Beyond required prediction, the project adds model-based fuel-type imputation, premium adequacy diagnostics, customer risk segmentation, and a stacking ensemble to connect model output to underwriting decisions.

1. Business Problem and Data

1.1 The Adverse Selection Problem

Motor insurance pricing must balance accuracy and competitiveness. Overcharging safe drivers pushes them to competitors; undercharging risky drivers leaves the insurer exposed. Accurate expected loss prediction is therefore the foundation of sustainable portfolio profitability.

The data are structurally challenging: 88.77% of policyholders made no claim, while non-zero losses are heavily right-skewed. This motivates Tweedie-based modeling, which handles both zero-loss probability and non-zero claim severity.

1.2 Target Variables and Engineered Features

Targets were derived from the training data: LC = X.15 / X.16 when X.16 > 0, else 0; HALC = LC x X.18; CS = 1 if X.16 > 0, else 0. The test set excludes X.15-X.18, so models must generalize from remaining variables.

Engineered variables include driver age, license years, policy duration, vehicle age, experience ratio, premium per experience, power-to-weight ratio, and cancellation rate (X.12 / X.8). These features translate raw policy dates, vehicle characteristics, and cancellation history into more meaningful risk signals.

2. Methodology

2.1 Data Cleaning and Model-Based Imputation

X.27 (fuel type) had 453 missing training values, all coinciding with impossible values in X.23 (horsepower = 0) and X.26 (doors = 0), indicating corrupt records. These 1.1% of rows were dropped. For 140 missing X.27 values in the test set, we trained a Random Forest classifier on vehicle features and imputed 130 Petrol and 10 Diesel. Compared with mode-fill, this preserved minority diesel cases and reduced distortion in downstream vehicle features.

2.2 Task 1 - Predicting LC and HALC

Four model families were benchmarked with Tweedie loss (power = 1.5), matching the zero-inflated and right-skewed structure of claim losses:

Model	LC Val MSE	HALC Val MSE	Notes
Tweedie GLM (Baseline)	324,175	1,026,099	Linear benchmark; too simple for nonlinear risk patterns
XGBoost Tweedie	317,710	1,018,744	5-fold CV tuned; strong nonlinear baseline
LightGBM Tweedie	317,852	1,008,615	Faster training; slightly stronger MSE
Regularized LightGBM	316,481	1,007,284	Strong regularized model
Stacking Ensemble	316,465	1,005,405	Best LC and HALC model; combines complementary learners

The Stacking Ensemble combines LightGBM, XGBoost, and Random Forest as base learners with a Ridge meta-model (5-fold CV), achieving the best LC validation MSE. For HALC, the Stacking Ensemble achieved the best result (MSE 1,005,405). A Two-Stage Model (classify claim occurrence first, then regress severity among claimers) was tested but underperformed: LC MSE rose 13.4% and HALC MSE rose 18.7%, mainly because classification errors compounded in Stage 1.

2.3 Task 2 - Predicting Claim Status

Model	Val ROC-AUC	Notes
Logistic Regression	0.6965	Linear baseline
Random Forest	0.7521	Nonlinear improvement
XGBoost (Final)	0.7884	Best selected model

XGBoost was selected with AUC 0.7884. Class imbalance was handled via scale_pos_weight around 8, with hyperparameters tuned using GridSearchCV.

3. Model Interpretation and Business Insights

3.1 SHAP Feature Importance

Figure 1 reports SHAP importance for the final loss-cost model. Policy duration dominates, followed by cancellation rate and net premium. This means that behavioral and contractual history carry more signals than basic vehicle characteristics such as fuel type, vehicle age, horsepower, and doors. Additional SHAP feature-importance outputs for HALC and Claim Status are provided in Appendix A1.

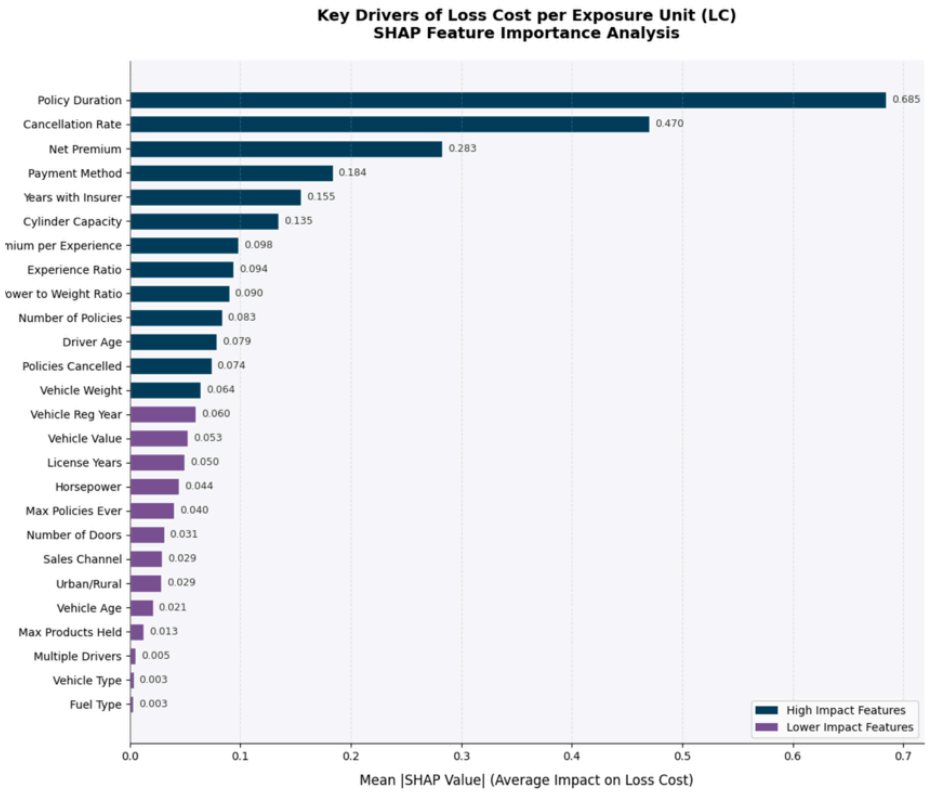


Figure 1. SHAP Feature Importance for LC - Key Drivers of Loss Cost per Exposure Unit

3.2 Driver Age Risk Segmentation

Age segmentation provides a directly interpretable business layer for underwriting. Young drivers (18-25) show the highest LC ($93.3), HALC ($199.0), and claim rate (15.3%). The 46-55 cohort shows a risk uptick (LC $66.0 vs. $60.0 for ages 36-45), while senior active drivers (56-65) are the safest group, with LC $49.4 and claim rate 9.6%.

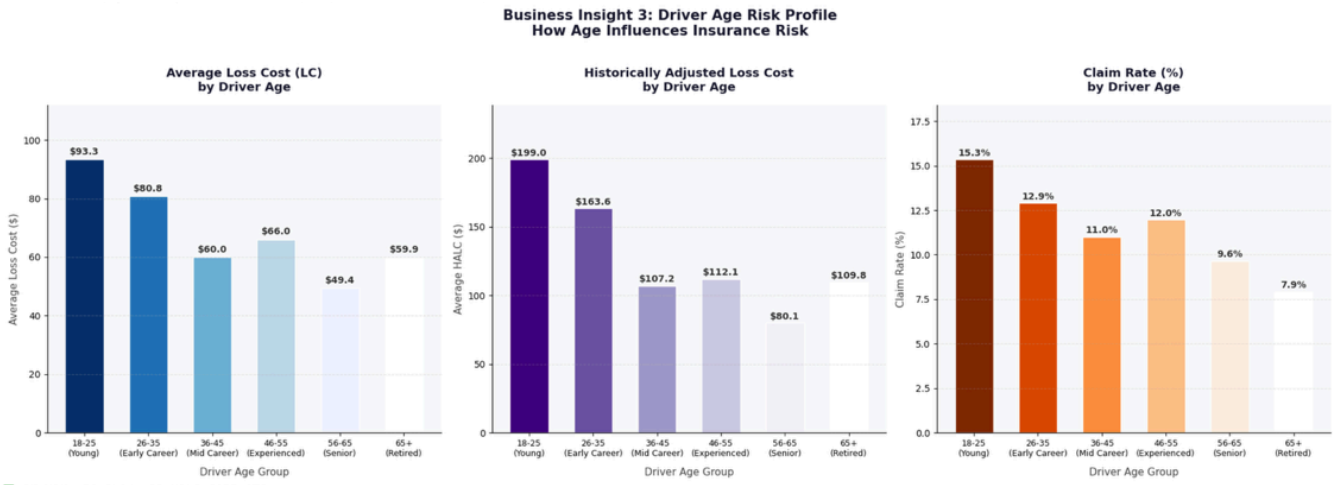


Figure 2. Average LC, HALC, and Claim Rate by Driver Age Group

3.3 Premium Adequacy Diagnostics

Premium adequacy diagnostics convert model output into a pricing question: are expected losses aligned with premium levels? LC/premium ratios remain below the 0.30 safe-zone benchmark across most premium segments, while HALC/premium ratios are higher for the lowest and medium premium bands. This suggests pure LC pricing appears broadly adequate, but historically adjusted risk remains more concentrated in lower-to-mid premium segments.

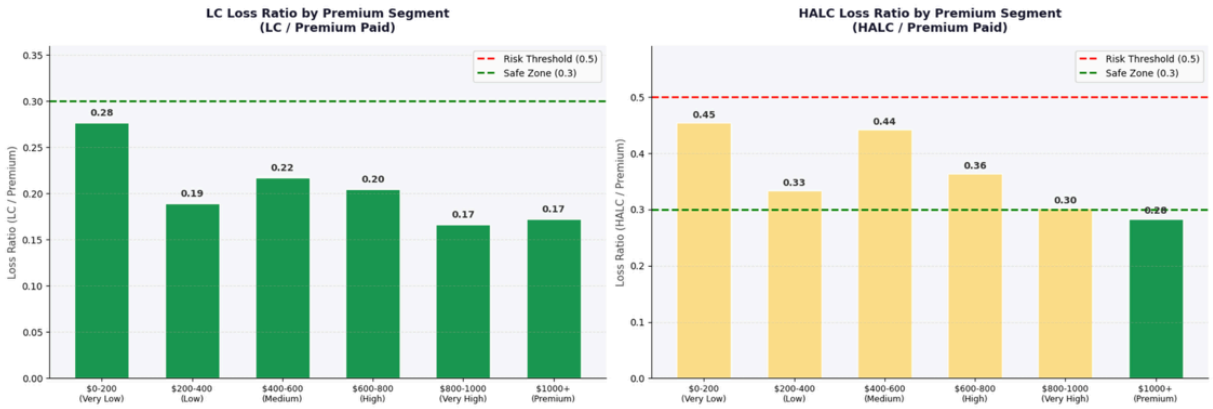


Figure 3. Premium Adequacy Analysis - LC and HALC Loss Ratios by Premium Segment

3.4 Customer Risk Segmentation

K-Means clustering separates policyholders into high-risk young drivers, medium-risk older drivers, low-risk low-premium customers, and loyal low-risk long-policy customers. This gives the insurer a business-facing segmentation layer for targeted underwriting, retention, and pricing actions, rather than relying only on individual policy predictions.

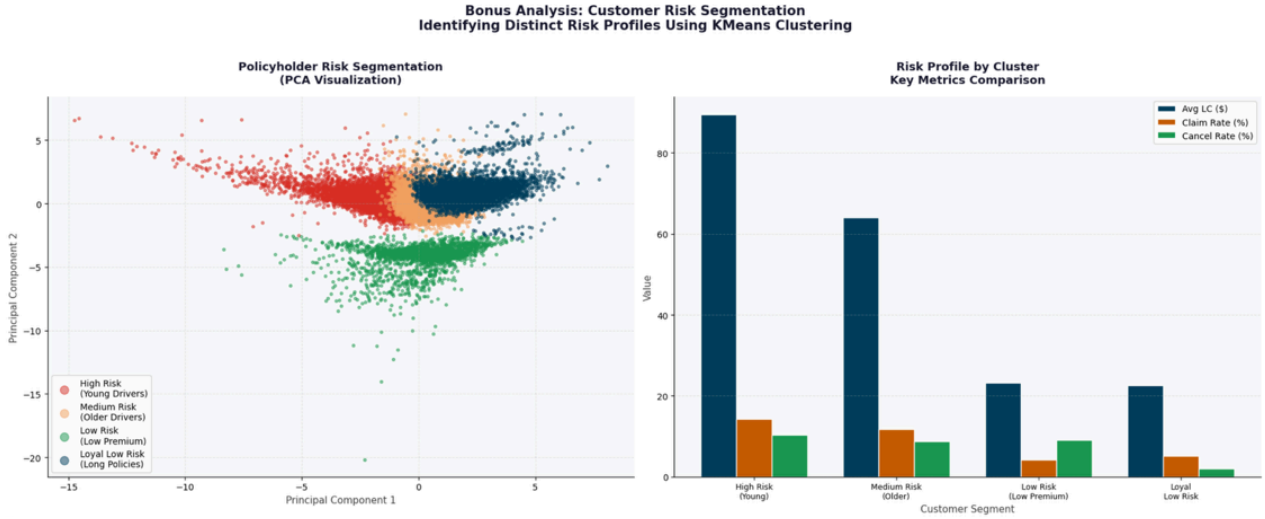


Figure 4. Customer Risk Segmentation Using K-Means Clustering

4. Business Understanding

Question	Business Answer
Accuracy vs. interpretation	Both matter. Validation performance drives financial outcomes; SHAP and segment diagnostics support governance, pricing justification, and business adoption.
Key ML challenges	Zero-inflated targets, heavy right tails, class imbalance, and temporal behavior features require Tweedie loss, imbalance handling, and careful validation.
Variable selection	All usable raw and engineered features were retained because tree-based models can exploit nonlinear interactions that univariate screening may miss.
Life insurance applicability	Not directly transferable: motor claims are short-horizon accidents, while life insurance depends on mortality, morbidity, and domain-specific actuarial tables.

5. Innovations and Recommendations

Innovation	Why It Matters
Model-based missing-value imputation	Predicting missing fuel type preserved minority diesel cases and avoided distortions from naive mode imputation.
Stacking ensemble	Combining LightGBM, XGBoost, and Random Forest achieved the best LC validation MSE and demonstrated ensemble diversity.
Premium adequacy analysis	Loss-ratio diagnostics connect predictions to pricing adequacy and identify segments where historical risk may be underpriced.
Risk segmentation	Age and cluster profiles translate model output into underwriting, retention, and targeted pricing strategies.

5.1 Actionable Recommendations

- Use policy duration as a renewal-risk signal: short-term policyholders should receive closer underwriting review, while long-term low-risk policyholders can be targeted with loyalty-based discounts or retention offers.
- Flag customers with very high cancellation rates for premium loading or manual review, while treating the 0–20% cancellation-rate group as the most stable segment.
- Tighten pricing and underwriting review for high-risk young drivers, especially drivers under 35; also investigate the 46–55 risk uptick using exposure, mileage, and professional-use information.
- Review vehicle-type pricing: vans show the highest claim risk and may require premium adjustment, while motorbike and agricultural vehicle segments should be checked for potential low-risk market opportunities.
- Focus premium adequacy review on the $0–200 and $400–600 premium bands, where HALC loss ratios are closest to the danger zone; these segments may need targeted 10–15% premium increases rather than broad portfolio-wide changes.
- Use the ensemble/Tweedie models as final prediction tools, but keep SHAP and segmentation diagnostics as explainability support for underwriting, pricing, and governance decisions.

5.2 Customer Risk Segmentation

Building on the cluster profiles in Section 3.4, each segment translates directly into differentiated underwriting and pricing actions. For Segment C0 (High Risk, Young Drivers), tighter premium loading and manual underwriting review are warranted given the 14.3% claim rate. Segment C1 (Medium Risk, Older Drivers) shows a potential underpricing gap — an 11.7% claim rate against an average premium of only $326 — and merits targeted adequacy review. Segment C2 (Low Risk, Low Premium) represents the most stable retention opportunity, with a 4.2% claim rate suitable for loyalty incentives or lower-touch renewal processes. Segment C3 (Loyal Low Risk) confirms that long-tenure policyholders remain the lowest-risk group and should be prioritized for retention offers. Applied alongside SHAP-based individual predictions, these segment profiles give underwriters a practical, interpretable layer for portfolio-level pricing decisions.

5.3 Deployment and Monitoring Plan

Area	Recommended Use	Monitoring / Guardrail
Pricing	Use LC and HALC predictions to flag segments where expected losses may exceed premium adequacy thresholds.	Track realized loss ratios by premium band and recalibrate if lower-to-mid bands remain high.
Underwriting	Prioritize policy duration, cancellation rate, and high-risk age profiles in manual review queues.	Avoid using a single SHAP feature as a hard rule; combine signals to prevent overreaction.

Retention	Identify loyal low-risk long-policy customers for retention offers or lower-touch renewal processes.	Monitor whether retention discounts preserve profitability rather than simply increasing exposure.
Governance	Use SHAP, segment charts, and benchmark tables to explain why final models were selected.	Maintain validation logs and compare future temporal holdout performance against current MSE/AUC.

5.4 Limitations and Future Work

First, validation is based on the available training split, so future performance should be checked with a temporal holdout if policy timing becomes available. Second, the premium adequacy and segmentation analyses are diagnostic rather than causal: high-risk segments should trigger review, not automatic pricing changes without actuarial and regulatory checks. Third, the two-stage model remains useful for frequency/severity interpretation even though it was not selected for final prediction. Future work could add threshold calibration for CS, exposure/mileage variables, and broader fairness review before production deployment.

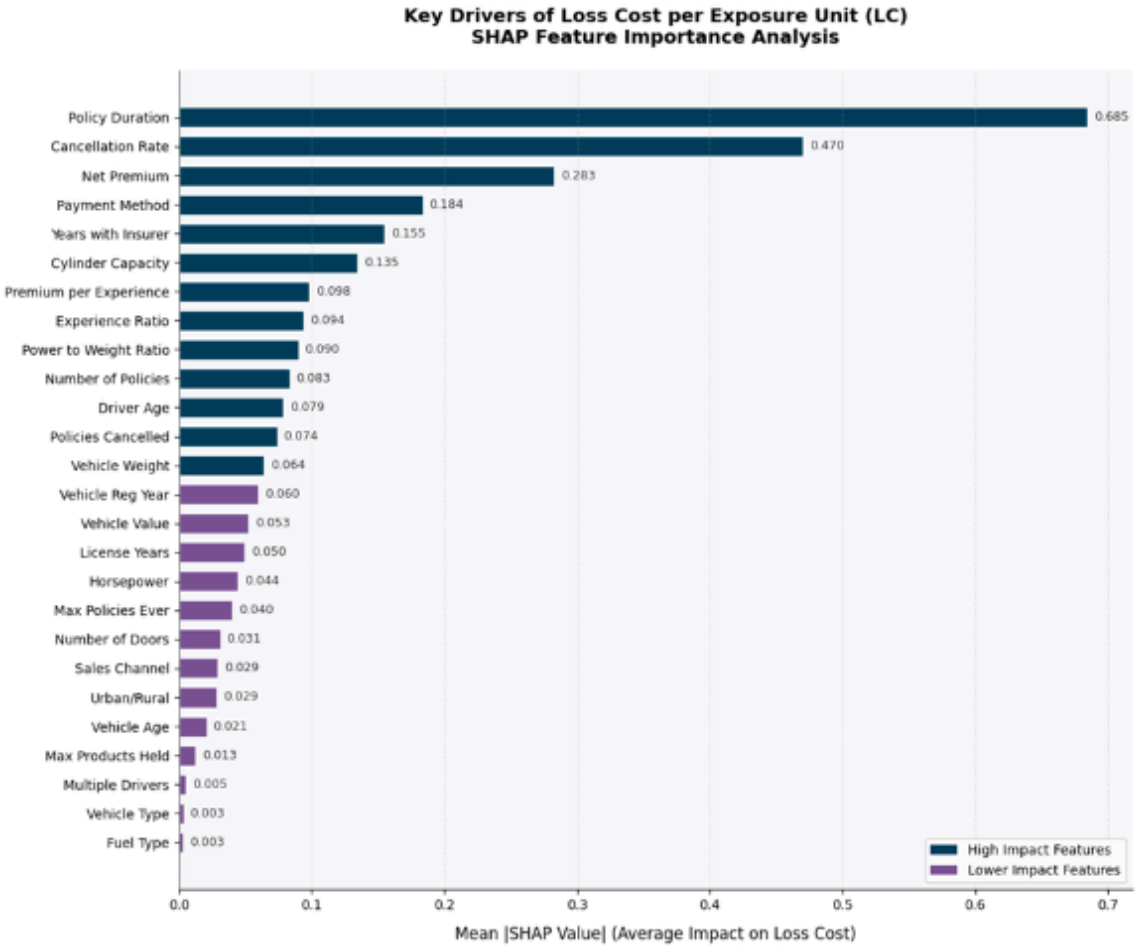
5.5 Conclusions

Modern gradient boosting with Tweedie loss substantially outperforms a traditional Tweedie GLM for insurance loss prediction while remaining interpretable via SHAP. Behavioral and contractual variables dominate vehicle characteristics for LC/HALC, while the XGBoost classifier provides meaningful claim discrimination despite class imbalance. The added imputation, premium adequacy, and segmentation analyses turn predictive modeling results into practical actuarial and business decisions: they improve data quality, connect predictions to pricing adequacy, and translate model outputs into segment-level actions.

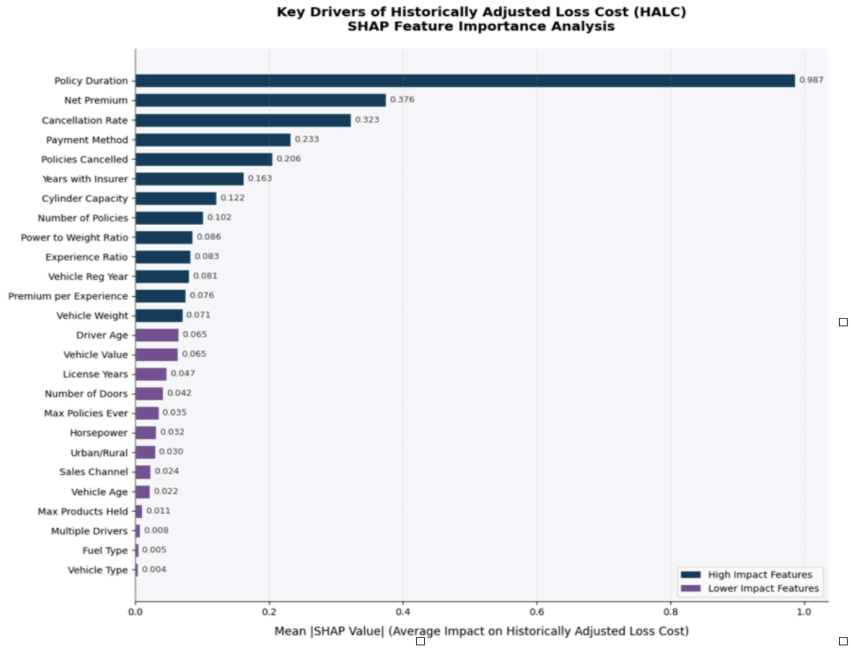
Appendix

A1. Model Interpretation (SHAP Analysis)

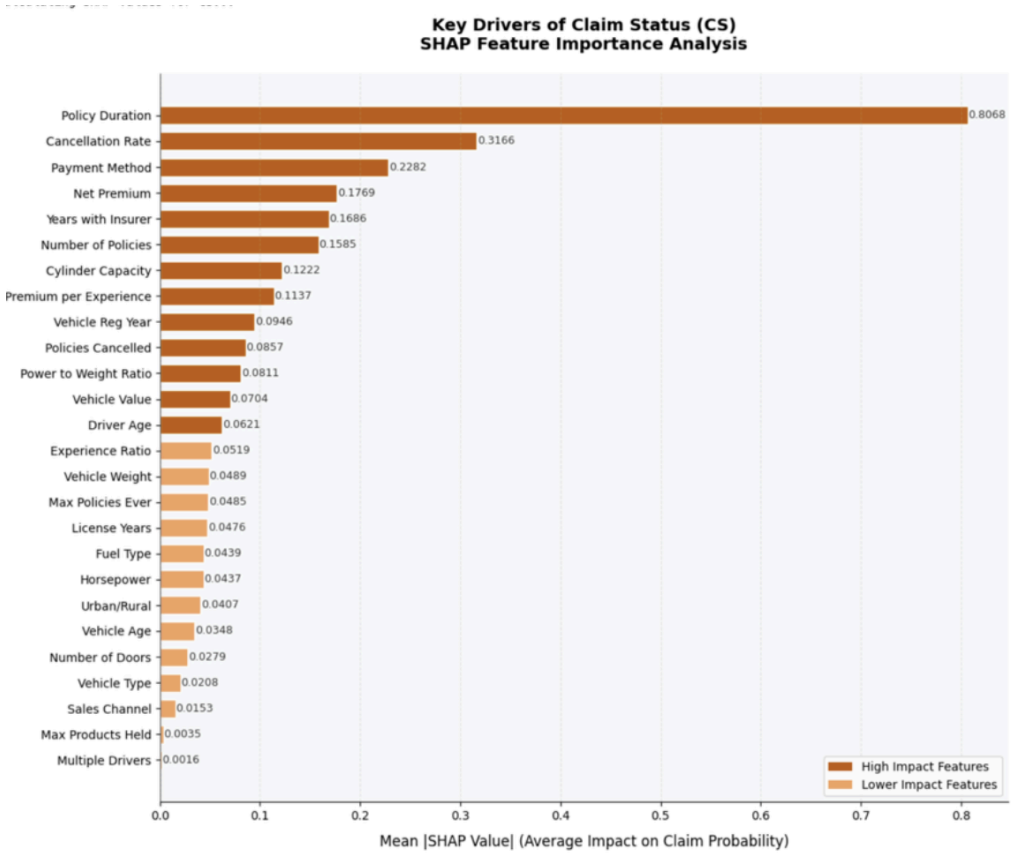
SHAP Feature Importance for Loss Cost Model



SHAP Feature Importance for HALC Model



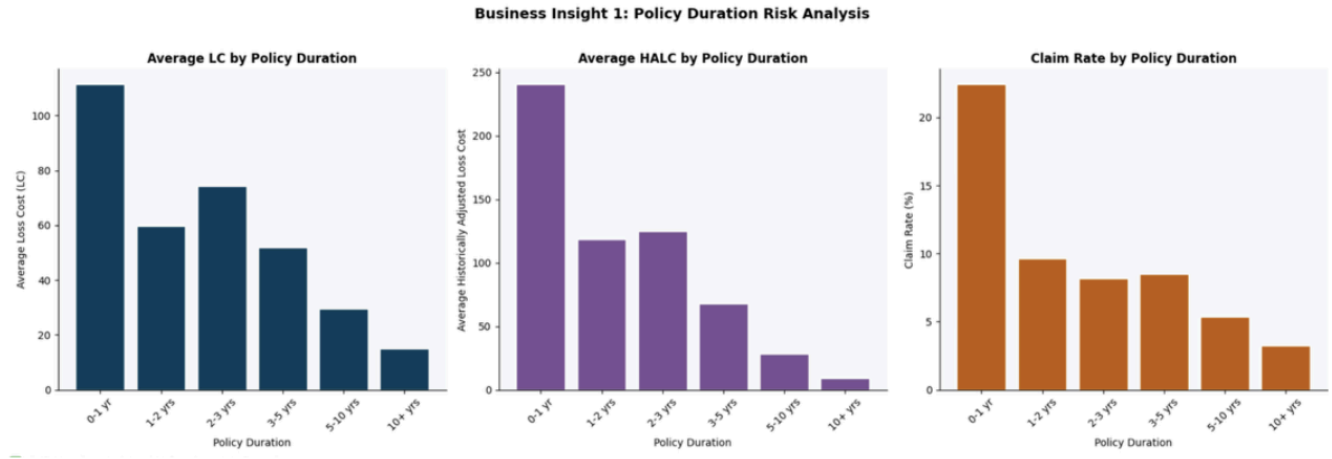
SHAP Feature Importance for Claim Status Model

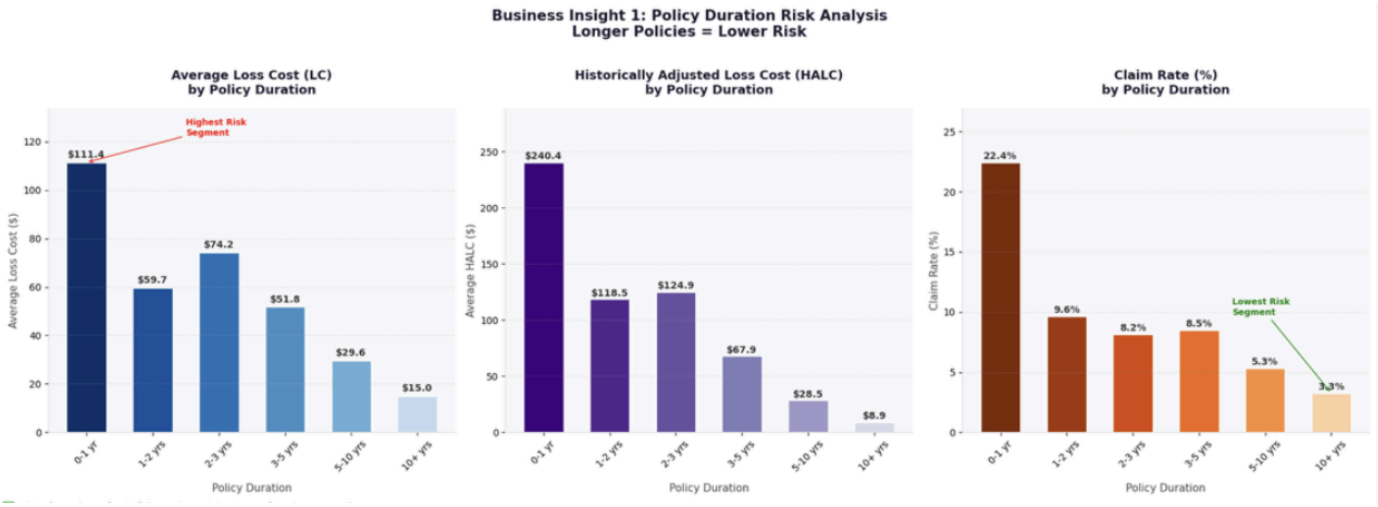


A2. Business Insights Visualizations

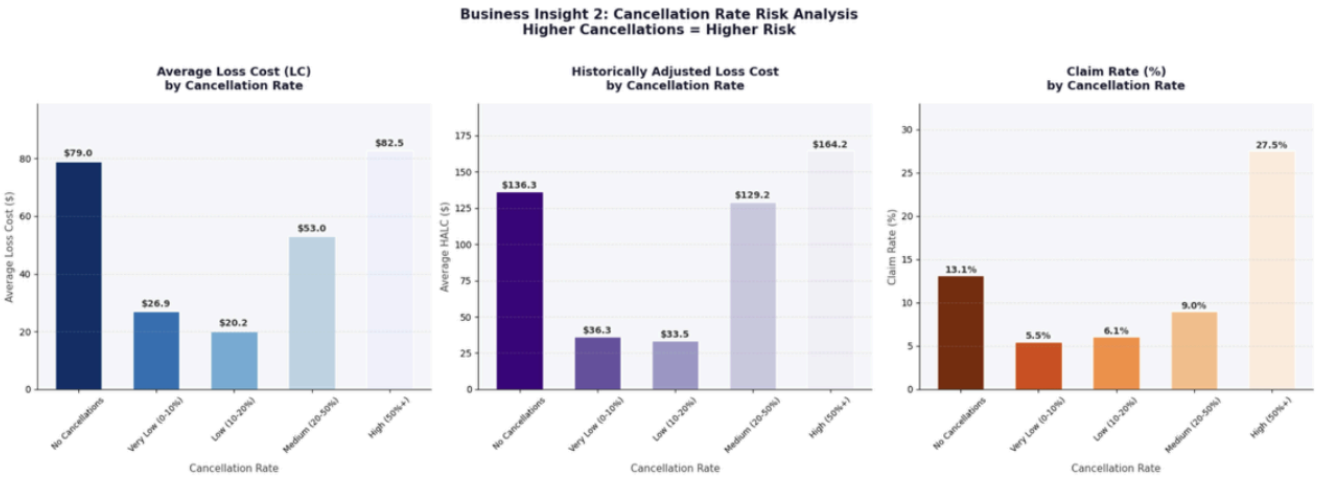
A2.1 Key Risk Drivers

Policy Duration vs Loss Cost



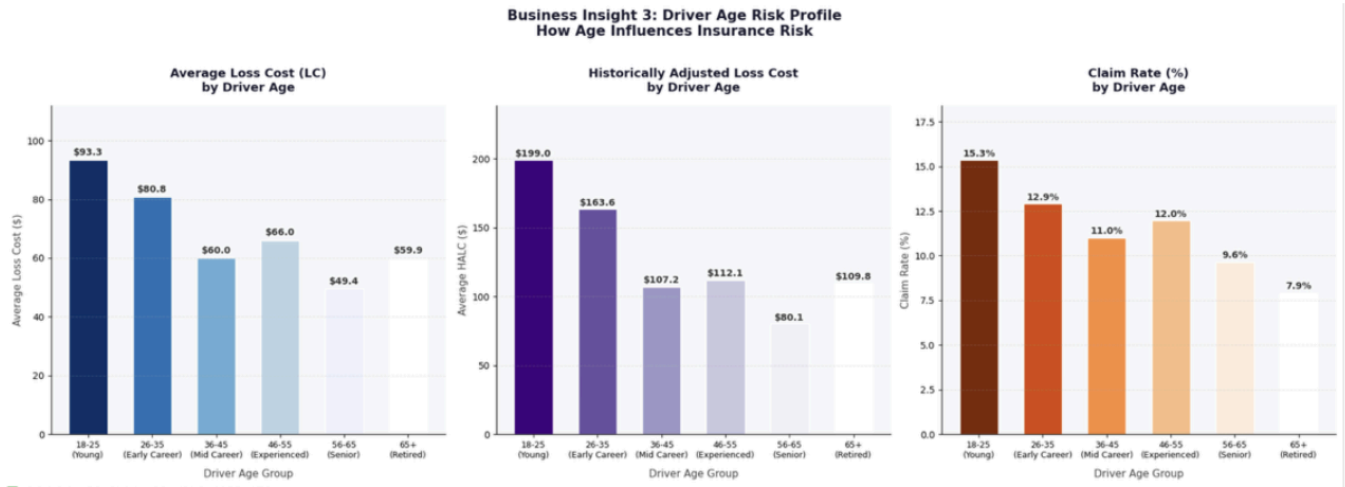


Cancellation Rate vs Loss Cost



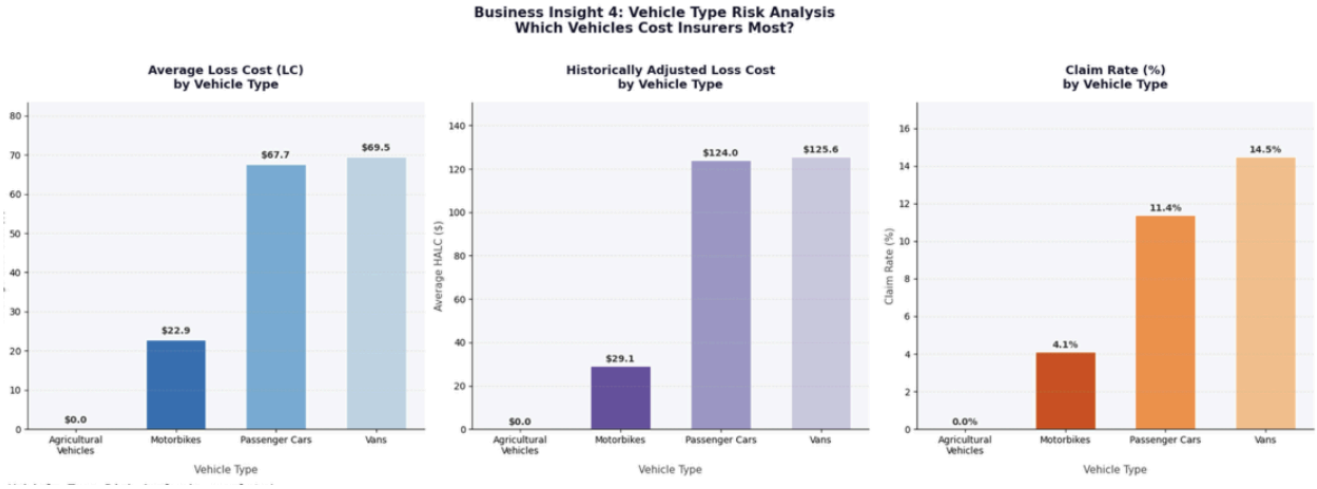
A2.2 Driver Age Analysis

Average LC, HALC, and Claim Rate by Driver Age Group

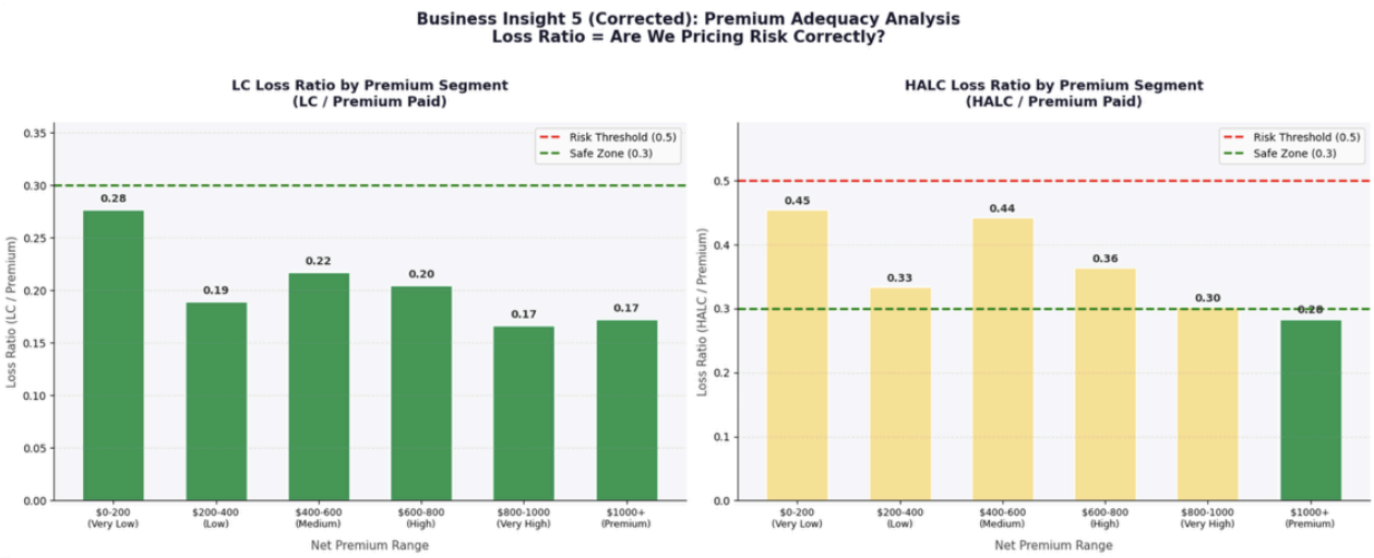


A2.3 Vehicle and Premium Analysis

Loss Cost by Vehicle Type

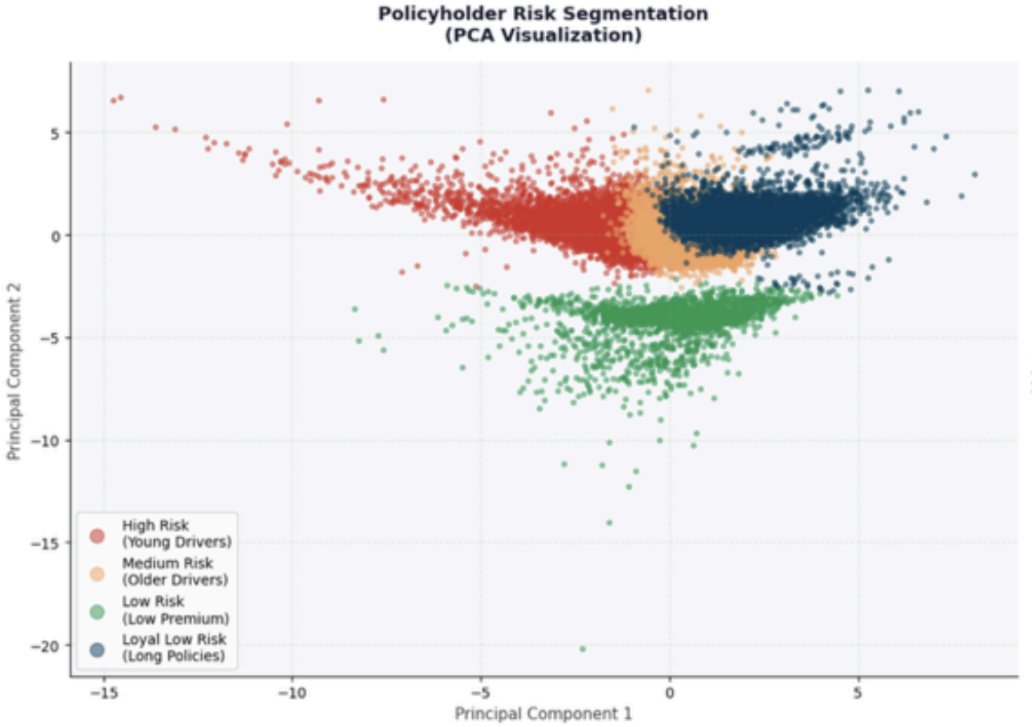


Premium Adequacy Analysis (LC and HALC Ratios)

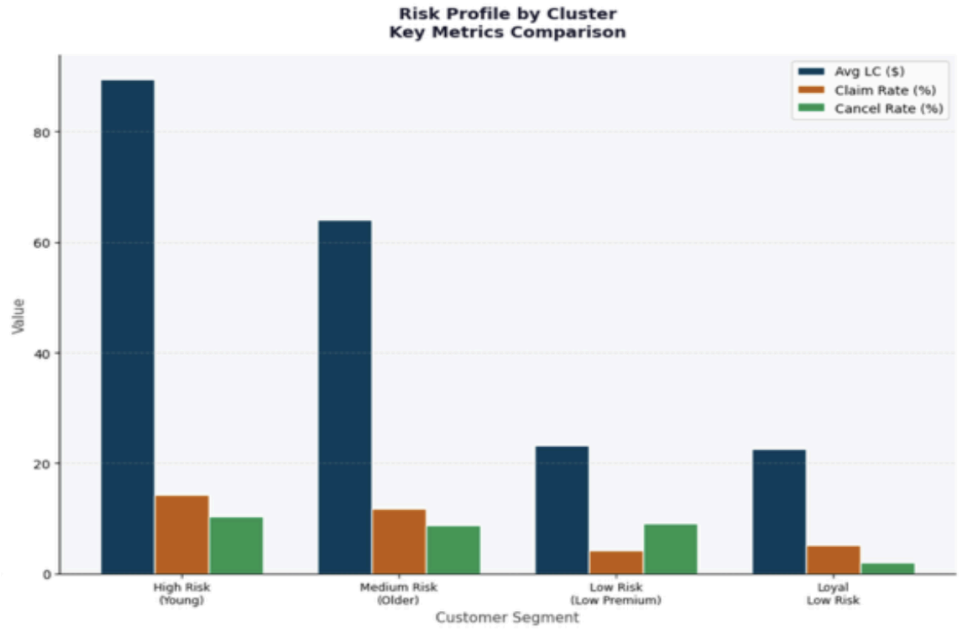


A3. Customer Segmentation Analysis

Customer Segmentation Using K-Means Clustering



Risk Profile Comparison Across Clusters



A4. Clustering Validation

Elbow Method for Optimal Number of Clusters

